

A singleton model is faithful and exactly compact-valued

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.faithful_and_compactOperatorModel_of_singleton`

theorem

Combines faithfulness with both inclusions needed to identify the represented range.

Statement

Let A be a nonzero complex C^* -algebra, with no unit assumed, and let $\pi : A \rightarrow \mathcal{B}(H)$ represent the unique unitary-equivalence class of nonzero irreducible $*$ -representations on a separable complex Hilbert space H . Then

$$\ker \pi = \{0\}, \quad \pi(A) = \mathcal{K}(H).$$

Equivalently, π is a faithful $*$ -representation and implements a $*$ -isomorphism $A \cong_* \mathcal{K}(H)$.

Assumptions

There is no assumption that A is separable, simple, or already faithfully represented. The singleton condition quantifies over nonzero irreducible comparison representations, without restricting them to separable spaces.

Conclusion

π is injective and satisfies the compact-operator model predicate: every $\pi(a)$ is compact and each compact operator has a preimage in A .

Proof route

Apply the three established results to the same singleton representation and assemble their outputs.

▼ Proof steps

1. Apply Faithfulness of a singleton representation to A and the given singleton π . Nontriviality and the usual C^* -algebra order are among the present assumptions; no separability is needed in that step. Its output is injectivity of this π .
2. Apply Every represented operator is compact to the same A, H, π . Its additional hypothesis is precisely the assumed separability of H . For every $a \in A$ it gives compactness of $\pi(a)$, hence $\pi(A) \subseteq \mathcal{K}(H)$.
3. For any compact $T : H \rightarrow H$, apply Every compact operator has a preimage using these same assumptions and that particular T . The result is $a \in A$ with $\pi(a) = T$, hence $\mathcal{K}(H) \subseteq \pi(A)$. Substituting all three outputs into The three clauses of a compact-operator model proves the model property, alongside the explicitly stated injectivity.

Main citations

- Exact theorem and proof
- Faithfulness of a singleton representation
- Every represented operator is compact
- Every compact operator has a preimage
- The three clauses of a compact-operator model

Lean source signature (exact)

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theorem faithful_and_compactOperatorModel_of_singleton [Nontrivial A]
  [TopologicalSpace.SeparableSpace H]
  (pi : NonUnitalCStarRepresentation A H)
  (hsingle : IsSingletonIrreducibleModel.{u, v, u} pi) :
  Function.Injective pi ∧ IsCompactOperatorModel pi
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In the source	Mathematical meaning
[Nontrivial A]	The algebra is nonzero: $A \neq \{0\}$. This condition does not say whether a unit is assumed; that information comes from the surrounding C^* -algebra structure.
[TopologicalSpace.SeparableSpace H]	The Hilbert representation space H is separable; A need not be.
(pi : NonUnitalCStarRepresentation A H)	The specified $*$ -representation $\pi : A \rightarrow \mathcal{B}(H)$; no unit-preservation equation is required.
(hsingle : IsSingletonIrreducibleModel.{u, v, u} pi)	The specified representation is nonzero irreducible, and every nonzero irreducible $*$ -representation of the same algebra is unitarily equivalent to it.
Function.Injective pi	Faithfulness of that same π .
∧ IsCompactOperatorModel pi	In addition, the same map is injective, sends every algebra element to a compact operator, and represents every compact operator on H .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration:

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